

Víctor Soria Pardos

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Summary

I am a Computer Architect specialized in multi-core architecture, hardware synchronization, cache-coherence, and memory systems. I currently work as a SoC Interconnect Architecture Engineer at OpenChip while serving as a Part-Time Teaching Assistant at Universitat Politècnica de Catalunya (UPC), where I earned my Ph.D. in Computer Architecture. I also a Master in Innovation and research and a Degree on Computer Science from the UPC and the Universidad of Zaragoza, respectively.

In my role at Openchip, I define hardware architecture specifications for AI and HPC accelerators. I assess, specify and develop CPU architectural solutions for Software and Hardware problems. I help to coordinate cross functional teams including staff from design, verification, physical design and emulation for different essential ASIC components. I also serve as a company representative in RISC-V Special Interest Groups (SIG), and collaboration with universities.

Previously, I spent over seven years at the Barcelona Supercomputing Center (BSC) researching on advanced hardware architectures, gem5 modeling, full-system hardware development, verification, and synthesis. Resulting in multiple top tier publications and open source CPU desings relevant in the european silicon ecosystem. I also mentored several students from different levels: degree, masters, and PhD.

Experience

Openchip, SoC Interconnect Architecture Engineer at HW Architecture Group 2026 - Now

- Working on the Hardware Architecture group, defining and researching cache coherence, hardware synchronization for chiplet-based AI accelerators based on RISC-V.
- Openchip representative in RISC-V Special Interest Groups: Datacenter and HPC.

Universitat Politècnica de Catalunya, Teaching Assistant 2022 - Now

- Teaching Computer Architecture I & II Labs (Computer Science degree 2nd and 3rd year).
- Technical Reviewer for Top Conferences including HPCA, MICRO, or ICCD.

Barcelona Supercomputing Center (BSC), Researcher at SIM Group 2018 - 2025

- Worked on the Simulation Infrastructure and Methodologies group (SIM), researching cache coherence, and hardware synchronization in gem5.
- Designed several domain-specific accelerators for HPC, machine learning, databases, and genomic analysis. Modeled in gem5 and implemented in SystemVerilog.
- Designed and taped-out two Academic CPUs (DVINO and Sargantana) with custom instructions to accelerate HPC, machine learning, and genomic analysis.

Arm Architecture and Technology Group, Sophia Antipolis, Research Internship 2023 - 2023

- Research and performance modeling of floating-point atomic RMW instructions. Included in the Armv9.6 ISA Extension as the Large System Float Extension (LSFE). Supervised by Eric Biscondi and Didier Martinot.

Education

Universitat Politècnica de Catalunya, PhD in Computer Architecture. Cum Laude 2021 - 2026

- Title: "Advanced Atomics: Leveraging Memory Hierarchies to Improve Atomic Memory Operations for Next Generation Coherent Manycores." Advised by Miquel Moretó (UPC/BSC, Spain) and Darío Suárez (Unizar, Spain)

Universitat Politècnica de Catalunya, Master in Innovation and Research in Informatics 2019 - 2022

- Minor in computer engineering for HPC, GPA: 3.7

Universidad de Zaragoza, Bachelor in Computer Science. GPA: 3.6 (Top 3/90) 2015 - 2019

Selected Peer-Reviewed Publications (4/14)

- [1] **V. Soria-Pardos**, A. Armejach, D. Suárez-Gracia, T. Mück, J. A. Joao, M. Moretó, "Delegato: Locality-Aware Atomic Memory Operations on Chiplets". In *Proceedings of the 58th Annual International Symposium on Microarchitecture. (MICRO '25)*. **Best Paper Candidate**.
- [2] M. Siracusa, **V. Soria-Pardos**, F. Sgherzi, J. Randall, D. J. Joseph, M. Moreto, A. Armejach, "A Tensor Marshaling Unit for Sparse Tensor Algebra on General-Purpose Processors". In *Proceedings of the 56th Annual International Symposium on Microarchitecture. (MICRO '23)*
- [3] **V. Soria-Pardos**, A. Armejach, D. Suárez-Gracia, T. Mück, J. A. Joao, A. Rico, M. Moretó, "DynAMO: Improving Parallelism Through Dynamic Placement of Atomic Memory Operations". In *Proceedings of the 50th Annual International Symposium on Computer Architecture. (ISCA '23)*
- [4] **V. Soria-Pardos**, M. Doblas, G. López-Paradís, G. Cándon, N. Rodas, X. Carril, P. Fontova, N. Leyva, S. Marco and M. Moretó, "Sargantana: A 1 GHz+ In-Order RISC-V Processor with SIMD Vector Extensions in 22nm FD-SOI". *Euromicro Conference on Digital Systems Design (DSD '22)*. **Outstanding Paper Award**.

Industrial and Academic Research Projects

Arm-BSC Center of Excellence	2020 - 2025
Designed and modeled new features for the AMBA CHI protocol in gem5.	
European Procesor Initiative (EPI)	2021 - 2025
Performance modeling (gem5) of Arm-based HPC multicore architectures	
PERTE-Chip Microelectronics & Semiconductors	2022 - 2024
Contributed to RTL design and integration of high-performance features in a multi-core system.	
DRAC Project	2019 - 2022
Designed and taped out two RISC-V CPUs with custom instructions for HPC and specific domain applications.	

Awards & Fellowships

MICRO'25 Best Paper Candidate	2025
Outstanding Paper Award DSD 2022	2022
HiPEAC Paper Award	2023, 2024, 2025
National Doctoral Fellowship (FPU'20) (Awarded to top 10% applicants)	2020

Technical Skills

Core Competencies: Microarchitecture, Cache Coherence, Interconnection, Synchronization, Domain-Specific Accelerator Design, High Performance Computing, Parallelism, Hardware/Software Co-Design.

Industrial Standards: AMBA, AXI, CHI, CHI C2C, RISC-V, Armv8/9-A, UCIe, CXL, UALink, PCIe.

Performance Modeling: Architectural modeling (gem5), HPC programming (OpenMP, MPI, CUDA), performance analysis tools (perf, VTune), low-level coding (Assembly, LLVM/GCC engineering, SIMD intrinsics).

Hardware Design: RTL (Verilog, SystemVerilog), Simulation (Verilator, Questasim), FPGA (Vivado), Synthesis (Synopsys DC, Cadence Genus), Silicon Tapeouts (TSMC 65nm, 2xTSMC 22nm, Intel 3), and Open-source Contributor (DVINO, Sargantana, gem5),

Software & Tools: C/C++, Python, Bash, Java, SQL, Docker, Latex, AsciiDoc, Confluence, Jira, Jama, Git.